



## 2.13

# Exponential and Logarithmic Equations and Inequalities

## Student Notes and Guided Practice

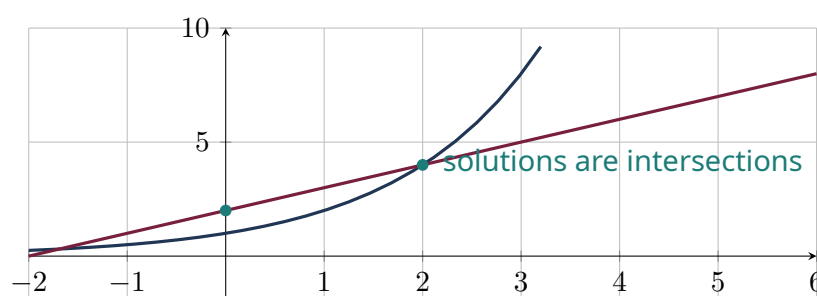
AP Precalculus - Unit 2 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

### Learning Targets

- Solve exponential and logarithmic equations exactly or approximately.
- Solve inequalities using monotonicity and sign reasoning.
- Reject extraneous logarithmic solutions.

### Essential Question

Which representation or property makes an exponential or logarithmic equation solvable?



## Concept Framework

### Key Idea

Use common bases when possible; otherwise apply logarithms.

### Key Idea

For logarithmic equations, state domain restrictions before solving.

**Key Idea**

An increasing function preserves inequality direction; a decreasing function reverses it.

**Key Idea**

Every proposed logarithmic solution must make all original arguments positive.

**Formula Map****Formula and Structure**

$$b^u = b^v \Rightarrow u = v$$

**Formula and Structure**

$$b^x = A \Rightarrow x = \frac{\ln A}{\ln b}$$

**Formula and Structure**

$$\log_b u = \log_b v \Rightarrow u = v \ (u, v > 0)$$

**Worked Examples****Worked Example 1**

**Problem.** Solve  $3^{2x-1} = 27$ .

**Solution.**  $27 = 3^3$ , so  $2x - 1 = 3$ , hence  $x = 2$ .

**Worked Example 2**

**Problem.** Solve  $2^x = 7$ .

**Solution.**  $x = \ln 7 / \ln 2$ .

**Worked Example 3**

**Problem.** Solve  $\ln(x - 1) + \ln(x + 1) = \ln 8$ .

**Solution.** Domain  $x > 1$ . Then  $x^2 - 1 = 8$ , so  $x = 3$  (reject  $-3$ ).

**Multiple-Representation Connection****AP Reasoning**

For this topic, always connect at least two representations: algebraic structure, numerical evidence, graphical behavior, or contextual meaning. State restrictions and units before interpreting a result. A correct calculation without a valid domain or contextual interpretation is incomplete AP reasoning.

## Common Errors to Avoid

### Common Error Check

- Do not discard domain restrictions when rewriting an expression.
- Do not infer local behavior from only one interval-wide average.
- Do not report a numerical answer without units or contextual meaning when quantities are named.
- Do not use a graphing window as proof that a feature does not exist.

## Guided Check

1. Solve  $2^{x+1} = 16$ , checking restrictions.

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2. Solve  $5^{2x} = 7$ , checking restrictions.

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3. Solve  $\log_3(x - 1) = 2$ , checking restrictions.

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4. Solve  $\ln x + \ln(x - 2) = \ln 3$ , checking restrictions.

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5. Solve  $4^x - 5 \cdot 2^x + 4 = 0$ , checking restrictions.

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6. Solve  $3^x < 20$  exactly as an interval.

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### Lesson Summary

Complete these sentences before leaving the lesson:

1. The most important structure in this topic is \_\_\_\_\_.
2. A restriction I must remember is \_\_\_\_\_.
3. One graphical feature I can justify algebraically is \_\_\_\_\_.